

1.3 Advantages & Contributions

1.3.1 Easy to access and achieve

The project provides a solution that does not require extra devices or complicated devices. It relies on the smartphones that are owned by the users. The system uses a **gamified pet-care interaction** that encourages regular use. By transforming the check-in into the act of 'caring' for a virtual cat, the system reduces the 'surveillance' feel of the app and replaces it with a sense of companionship and purpose.

1.3.2 Reducing the unfortunate cases & Notified families in time

EthernaCare contributes by providing a structured process for identifying prolonged user inactivity and attempting to notify the appropriate person. When a configured inactivity window is missed, the application first reminds the user locally. Continued inactivity causes the system to attempt an SMS to the user's verified phone number before escalating to the selected trusted contact. When location permission is granted and a valid device location is available, a Google Maps link may also be included in the emergency message. This process may help family members or trusted contacts respond earlier to a possible emergency or delayed death discovery. However, successful notification depends on internet connectivity, external service availability, mobile carrier delivery and correct contact information.

1.4 Project Plan

The project follows an Agile mobile application development approach. Development is divided into several phases that include planning, system design, implementation, testing, and deployment. Core features such as activity detection and emergency alert notification are developed early in the project. Features related to funeral preferences, will management, and AI-assisted guidance are developed in later stages. Testing is carried out throughout the development process to ensure that each function works correctly and meets user needs.

Project Schedule:		
ACTIVITIES	EXPECTED OUTCOME	COMPLETION DATE
Proposal	Determine the objective, problems and solutions.	16/12/2025
Phase 2: FYP1		
Chapter 1: Introduction	Introduce the project purpose, background, objective and justification	26/1/2026
Chapter 2: Research background	Retrieve the relevant information, system requirements and research gaps	7/2/2026
Chapter 3: Methodology	Defined the system development tools and requirements.	24/2/2026
Chapter 4: System Design	Mock up prototype, system architecture and data flow diagram	14/3/2026
Project I Portfolio	Submission	13/4/2026
Phase 3: FYP2		
Chapter 5: Development & Testing	Develop the main functions	26/6/2026
Chapter 6: System deployment	Deploy and test the system in real-world scenarios.	6/7/2026
Chapter 7: Project conclusion	Final summary and outcome of the project	1/8/2026

Figure 1.4: The project schedule for each phases

1.5 Project Team & Organization

This project is developed as an individual final year project under the guidance of a project supervisor. The student is responsible for system design, development, testing, and documentation. The supervisor provides academic guidance, technical advice, and feedback to ensure that the project meets academic requirements and follows proper development practices.

1.6 Chapter Summary and Evaluation

This chapter introduced Ethernacare and explained the social problem associated with older adults and other individuals living alone. The system uses Oren, a virtual companion, to make safety check-ins more engaging and less clinical. Users can configure a rolling inactivity threshold from one to 168 hours, while staged reminders and SMS escalation help identify prolonged inactivity. In addition to check-in monitoring, Ethernacare provides Oren energy management, tokens, toys, virtual rewards, trusted contacts, emergency location support, AI-assisted guidance and protected Legacy Planning. The project is designed as a safety-support tool and does not replace medical monitoring, emergency services or direct family communication.

Chapter 2

Literature Review

2 Literature Review

This chapter examines previous research and existing systems related to elderly individuals who live alone and the growing concern of lonely deaths. It looks at demographic changes such as population ageing and the rise in independent living among older adults. It also reviews current technology solutions, including wearable emergency devices, smart home monitoring systems, and mobile safety applications. Through this review, the chapter highlights the strengths and weaknesses of existing approaches and clarifies how the proposed project addresses identified gaps and limitations.

2.1 Aging Population in Malaysia

2.1.1 Demographic Shift Toward an Ageing Society

Malaysia is moving toward an ageing nation as the percentage of citizens aged 60 and above continues to increase. The Department of Statistics Malaysia (*Current Population Estimates, Malaysia, 2023*, 2023) projects that older adults will make up around **15%** of the population by 2030. This change is mainly influenced by longer life expectancy and lower birth rates. The transition creates new challenges in healthcare planning, elderly care services, and community support systems.

2.1.2 Increase in Elderly Living Alone

The increase in elderly individuals living alone is closely linked to broader demographic and social transitions. According to the *World Report on Ageing and Health* by the World Health Organization (2015), although more generations within a family may now be alive at the same time due to increased life expectancy, these generations are increasingly living separately. In many countries, the proportion of older people living alone is rising dramatically. For example, in several European countries, more than 40% of women aged 65 and above live alone. Even in societies with strong traditions of multigenerational living, such as Japan and India, extended family households are becoming less common. Declining fertility rates mean there are fewer younger family members available to provide care, while changing gender roles have reduced the traditional caregiving capacity within families. Smaller household sizes may limit reciprocal care arrangements and increase the risk of poverty. Furthermore, older adults living alone may face higher risks of social isolation and suicide. Although many older people prefer to remain in their own homes and communities for as long as possible, these structural changes suggest

that traditional family-based care models are becoming less sustainable in the modern era (World Health Organization, 2015).

2.2 Technology concepts

2.2.1 Smartphone-Based Passive Monitoring

Smartphone-based monitoring systems use built-in sensors and activity logs to observe user behavior without requiring constant manual input (Wang, 2014). This passive approach allows the system to detect unusual patterns such as prolonged inactivity, which may indicate potential emergencies. Because the monitoring process runs automatically in the background, it reduces the need for continuous interaction and can operate without disturbing users.

2.2.2 Daily Check-In Monitoring Systems

Daily check-in monitoring systems are used in safety and well-being applications to verify that users are active through simple interactions with a mobile device. Instead of continuously collecting sensor data, gamified check-in systems utilize **virtual pets or interactive elements** to maintain user engagement. By requiring a tactile interaction like 'petting' an on-screen character, the system ensures the user is physically capable and responsive, serving as a more interactive form of passive monitoring (Rashidi & Mihailidis, 2013).

2.2.3 Wearable or Sensor Monitoring

Wearable and sensor-based monitoring systems are commonly used in elderly care technologies to continuously track health conditions and daily activities. These systems typically involve wearable devices such as smartwatches or sensor bands that collect physiological and movement data, including heart rate, walking patterns, and sleep activity. The collected data can be analyzed to identify abnormal patterns such as falls, prolonged inactivity, or sudden changes in physical behaviour (Patel et al., 2012).

2.2.4 Duolingo and Behavioral Gamification

Modern gamification success stories, most notably **Duolingo**, demonstrate that habit formation is driven by emotional engagement and loss aversion. According to research on Duolingo's growth strategy, the use of a persistent mascot (Duo the Owl) and a daily "streak" mechanic transforms a high-effort task into a "sticky" daily routine (Hasan, 2026). By quantifying

consistency, users feel a sense of ownership over their progress—a psychological phenomenon known as the **Endowment Effect**.

EthernaCare adapts these proven tactics by replacing standard check-in prompts with a virtual cat companion. By leveraging the emotional bond between the user and the digital pet, the system ensures high retention rates, as users are motivated to "check-in" to maintain their pet's well-being and their own reward progress.

2.3 Technology Acceptance Among the Elderly

2.3.1 Smartphone Accessibility Among Elderly Users

In recent years, smartphone ownership among older adults has increased significantly. According to the Pew Research Center (2021), smartphone adoption among individuals aged 65 and above has grown substantially compared to previous decades. The increased accessibility of smartphones provides opportunities to develop mobile-based solutions for elderly safety and monitoring. Since many elderly individuals already own smartphones, mobile applications can serve as cost-effective tools without requiring additional hardware (Pew Research Center, 2025).

2.3.2 Perceived Usefulness and Ease of Use

Recent studies show that perceived usefulness and ease of use remain strong predictors of technology adoption among older adults. Research on digital health and mobile monitoring systems indicates that elderly users are more likely to adopt applications when they believe the system improves their safety and supports independent living (Anderson et al., 2020). Confidence in smartphone usage and prior experience with digital tools also influence acceptance. When applications are perceived as complicated or unnecessary, adoption rates decrease.

2.3.3 Importance of User-Friendly Design

Usability plays a critical role in elderly technology adoption. Studies on mobile health applications emphasize that simple navigation, readable fonts, clear instructions, and minimal interaction steps significantly improve user satisfaction and long-term engagement (Dusseljee-Peute et al., 2018). Complex menus and small interface elements may discourage elderly users. Therefore, systems designed for older adults should prioritize clarity, accessibility, and reduced cognitive effort.

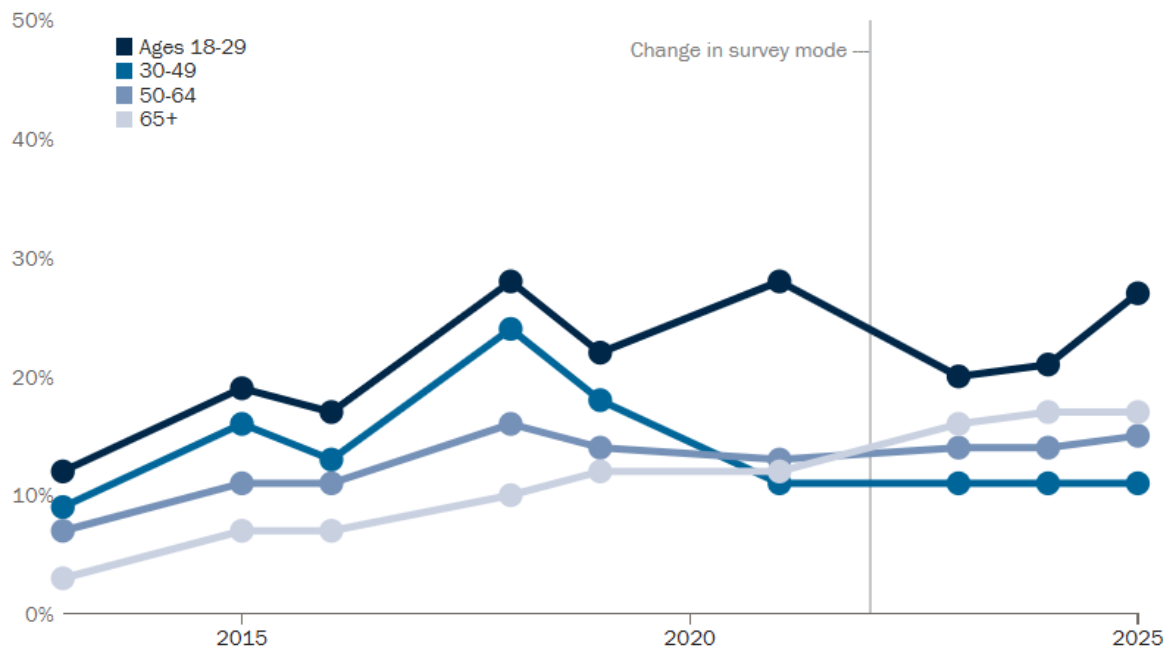


Figure 2.1: Percentages of U.S. adults who are smartphone dependent, by age

(Pew Research Center, 2025)

2.4 Similar Existing Systems in the market

2.4.1 Apple Emergency SOS

Apple Emergency SOS is a built-in smartphone feature that allows users to quickly contact emergency services and notify selected emergency contacts. The system can be activated using hardware button combinations, and it automatically shares the user's location with emergency responders and trusted contacts (Apple Inc., n.d.). The main feature is immediate emergency calling with real-time location transmission. It is integrated into iOS devices and depends on manual activation by the user.

This system is related to Ethernacare because both involve emergency alerts and contact notification. Ethernacare differs by using a configurable rolling check-in threshold and staged user/contact SMS escalation rather than direct emergency-service dispatch. It also includes protected Legacy Planning, which is outside Apple Emergency SOS.

2.4.2 CarePredict

CarePredict is a wearable-based monitoring system designed for elderly individuals, particularly those in assisted living facilities. The system uses a wearable device equipped with sensors to monitor daily activities such as eating, sleeping, walking, and bathroom visits. It applies artificial intelligence and behavior pattern analysis to detect anomalies that may indicate health or safety risks (CarePredict, n.d.).

The main feature is continuous activity tracking through wearable sensors combined with AI-based behavioral analytics. The strength of the system lies in its detailed monitoring capabilities. However, it requires users to wear a dedicated device and is often implemented in institutional care settings.

Ethernacare shares the objective of reducing delayed awareness but does not perform passive medical or behavioural sensing. It relies on an explicit check-in, scheduled threshold evaluation and trusted-contact escalation, without requiring a wearable device.

2.5 Limitation of the project

2.5.1 No Direct Medical Monitoring

Ethernacare does not measure heart rate, blood pressure, oxygen level, body temperature or other medical information. It does not include automatic fall detection and cannot diagnose illnesses or determine the cause of a user's inactivity. A successful Oren interaction only confirms that a check-in was recorded by the application. Therefore, Ethernacare must not be treated as a medical monitoring or diagnostic device.

2.5.2 No Automatic Emergency-Service Dispatch

The application does not automatically contact hospitals, ambulances, police or other Malaysian emergency services. Its automated escalation process attempts to send messages to the verified user and the selected trusted contact. The user may manually contact emergency services when necessary, while the trusted contact remains responsible for deciding what further action should be taken.

2.5.3 External Provider and Internet Dependency

Several functions depend on internet connectivity and third-party providers. SMS delivery depends on the SMS provider and the receiving mobile carrier, while email delivery depends on the email provider and recipient server. AI guidance, server-side processing and cloud database functions also require internet access. Therefore, successful request processing does not always prove that a message was received by its intended recipient.

2.6 Chapter Summary and Evaluation

This chapter analysed the challenges of social isolation and delayed awareness of emergencies among people who live alone. The reviewed systems support the value of accessible safety signals and trusted-contact communication. Ethernacare therefore uses a deliberate Oren interaction as a non-medical check-in, combines it with configurable threshold monitoring, and clearly limits its role to support and notification rather than diagnosis, fall detection or guaranteed emergency response.